10 Light and Shadows

Light enables us to see the objects around us. These objects may or may not allow light to pass through them. When light cannot pass through an object, it sometimes forms a shadow of the object. We will learn about light and shadow in this chapter.

Learn about

- Light
- How do we see things?
- Shadow
- Formation of shadow at different times of the day
- Movements of Earth
- Formation of day and night
- Eclipses

Light

We see the things around us with the help of light. Light is a form of energy. *Any object that produces light is called a **source of light**.* Sources of light are of two types: natural and artificial.

Fact File

Optics is the study of light.

*The sources of light that give out light on their own in nature are called **natural sources of light**.* The sun is a natural source of light. Other examples of natural sources of light are stars, lightning, and fires due to natural causes. A firefly also gives out light, so it can also be considered a natural source of light. The sun is the main source of light on Earth.

*The sources of light that are made by human beings such as electric bulbs, fluorescent tubes, torches, and candles are called **artificial sources of light**.*

Natural sources of light

Artificial sources of light

Sources of light

Light Travels in a Straight Line

An important property of light is that it travels in a straight line. When the path of light is blocked by an object, it causes the direction of light to change. However, light cannot bend around objects. If light could bend, we would be able to see things behind a high brick wall. We would also be able to see inside all the rooms of our home while standing downstairs.

Fact File

Light travels at a speed of about 300,000 km/s.

Activity

SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Aim: To show that light travels in a straight line

Materials required: A long, straight hollow pipe or straw, and a candle

Procedure: [**Note: Adult supervision required**]

1. Hold the pipe or straw in the front of the candle flame such that one end of the pipe or straw is facing it. Keep a safe distance between the pipe (or straw) and the flame.
2. Look at the flame through the open end of the pipe or straw (Fig. A).
3. Now, bend the pipe or straw and look through it (Fig. B). Observe what happens.

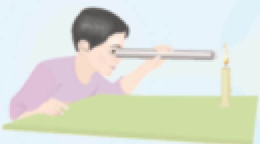


Fig. A

Fig. B

Observation: When the straw or tube is straight, we can see the light. When the pipe or straw is bent, we cannot see the light.

Conclusion: Light travels in a straight line.

Can We See Through Things?

We can see through clean water or glass, but not through a book or a brick wall. This happens because different materials allow different amounts of light to pass through them.

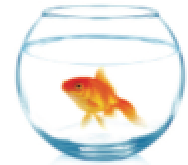
Materials such as wood and cardboard do not allow any light to pass through them. We cannot see through these objects at all. Such materials are said to be **opaque**.

Materials such as tracing paper and coloured glass allow only some light to pass through them. We can see, but not clearly, through these objects. Such materials are said to be **translucent**.

Materials such as clear water, cellophane, and glass allow light to pass through them completely. We can see very clearly through these objects. Such materials are said to be **transparent**.

Cardboard (Opaque)

Coloured glass (Translucent)



Glass (Transparent)

Activity

Aim: To identify transparent, translucent, and opaque objects

Materials required: Objects such as a pencil, a block of wood, a mirror, cellophane paper, a steel glass, tracing paper, a stone, water in a glass bowl, a piece of thin cloth, etc., and a source of light (such as a torch or an electric bulb)

Procedure:

1. Take each object one by one. Hold it against the source of light.
2. Identify the object as transparent, translucent, or opaque and write in the following table.

Observation and conclusion:

Transparent	Translucent	Opaque

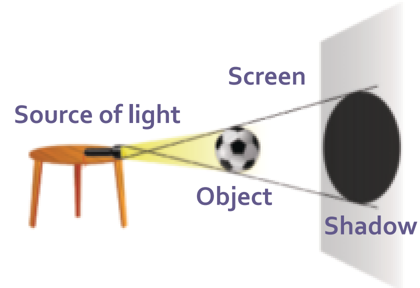
Shadow

We have learnt that light travels in a straight line. *When an opaque object comes in front of the light, it blocks the light and forms a dark patch called a **shadow**.* A shadow is always formed on the opposite side of the light. When we block sunlight, the shadow of our body is formed. When we move, our shadow moves along with us. [**Note:** It is not essential for it to form a shadow each time. There are certain conditions for shadow formation]. Opaque objects form dark shadows because they do not allow any light to pass through them. Translucent objects allow some light to pass through, so they form very faint shadows. Transparent objects allow light to pass through them completely, so they do not form a shadow at all.

Conditions Required for the Formation of Shadows

For the formation of a shadow to take place, we need to have the following three things:

i. A source of light
ii. An opaque object
iii. A surface (such as a wall or a screen) on which the shadow would form



Formation of a shadow

Characteristics of Shadows

The following are some characteristics of shadows:

- Shadows are formed only when a source of light is present.
- Shadows are always formed on the opposite side of the source of light.
- A shadow is always black in colour. It does not show the colour of the object. In dim light, the shadow formed will be light and faint. But in bright light, the shadow will be darker and easier to see. During the night, if there is total darkness (as there is no source of light) no shadow will be formed.
- A shadow does not show the details of an object. It only shows the outline.
- The size and clarity of the shadow depend on the following:
  i. the size of the object and the source of light
  ii. distance between the object and the source
  iii. distance between the object and the screen

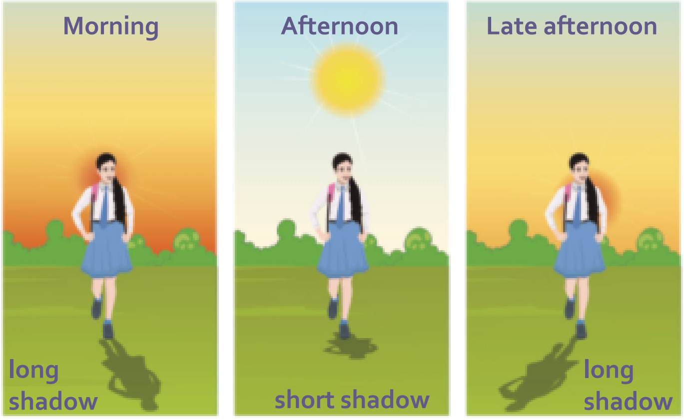
Think and Discuss

21st CENTURY SKILLS

Can you think of examples where you have seen shadows change in size or darkness?

Formation of Shadow at Different Times of the Day

When we step outside in the sunlight, we can see our shadow and also of several objects around us. The length and position of our shadow in sunlight are different at different times of the day. The sun appears to move across the sky from morning to evening. Accordingly, the shape and length of the shadows change as well. Shadows are the longest in the morning and in the late afternoon. They are the shortest at noon when the sun is overhead.



Shadows at different times of the day

Questions

Fill in the blanks.

1. An important property of light is that it travels in a .... line.
2. Light cannot .... around objects.
3. Two examples of translucent objects are .... and ....
4. For the formation of shadow, we need ...., ..., and ....
5. Shadows are the longest in the .... and in the ....

Activity

Aim: To study the formation of a shadow and its characteristics

Materials required: An electric bulb and a pen

Procedure:

1. Place the pen between the bulb and a wall (which acts as the screen).
2. Move the pen towards and then away from the bulb.

Observation: A dark shadow is formed on the wall. The shadow becomes bigger when the object is moved closer to the source of light (Fig. A). It becomes smaller when the object is moved away from the source of light (Fig. B).

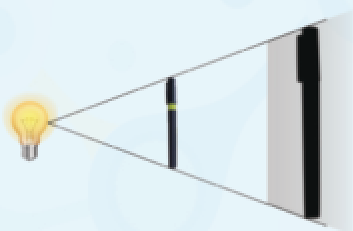


Fig. A

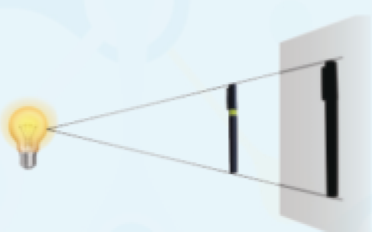


Fig. B

Conclusion:

1. Shadow is formed on the opposite side of the source.
2. Shadow is black in colour, and shows only the outline of the object.
3. The size and clarity of the shadow depends on the distance between the source and the object, and between the object and the screen.

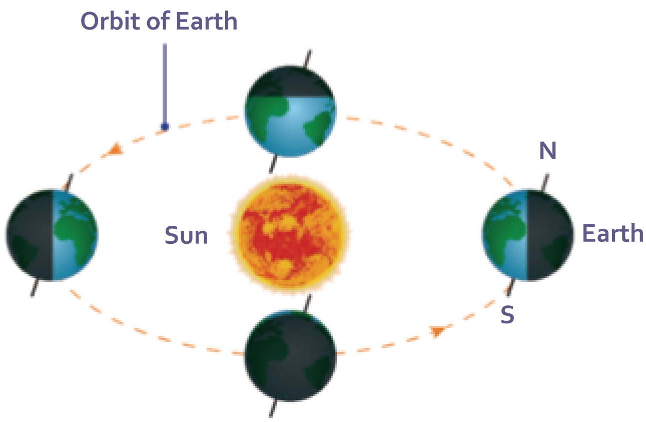
Movements of Earth

*Earth is a planet that moves around the sun in a fixed path called an **orbit**.*

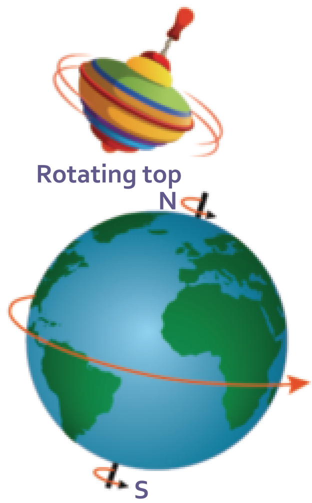
Earth shows two types of movements: revolution and rotation.

Revolution

The movement of Earth around the sun in its orbit is called a ***revolution****.* The orbit of Earth is an imaginary path. Earth takes 365 days and six hours to complete one revolution. The revolution of Earth is important because it plays an important role in the change of seasons.



Revolution of the Earth



Rotation of the Earth

Rotation

The Earth spins around its axis. Earth's axis is an imaginary line that passes through its centre. *The spinning movement of the Earth about its axis is called* ***rotation****.* Earth takes about 24 hours to complete one rotation. The rotation of Earth is important as it causes the formation of day and night. To see what the rotating Earth looks like, look at a spinning top. You will see the top spinning around its own axis.

SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Activity

Aim: To make a sundial (When there were no clocks or calendars, sundials were used to tell the time by looking at the length and direction of the shadow of the sundial in sunlight.)

Materials required: A clean, empty flower pot (no soil) with a hole at its bottom, a long stick, and a black permanent marker pen

Procedure:

1. Place the flower pot in an inverted position. Fix the stick into the hole at the bottom of the flower pot.
2. Now place the flower pot in an open place in the sunlight.
3. Notice that the shadow of the stick moves along the base of the pot.
4. Using the black marker, mark and also number the position of the shadow at the base of the pot every hour, from morning till the evening.
5. The sundial is ready! Any day, when the sun is visible in the sky, you can tell the time by observing the position of the stick's shadow on the marked numbers.

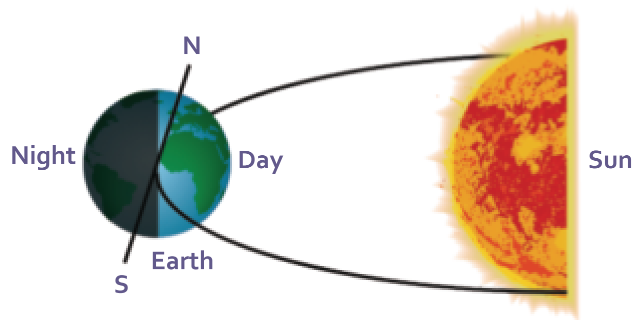

A sundial

Observation: The sun appears to move across the sky in a semicircle. So, the position of the stick's shadow also changes each hour.

Conclusion: Change in position of the shadow is due to the rotation of the Earth.

Formation of Day and Night

The rotation of Earth causes the formation of day and night. As Earth rotates, one half of it faces the sun. This part of Earth gets light from the sun and has day. The other half of Earth faces away from the sun and has night. India and the United States of America (USA) are located in the two opposite halves of Earth. So, when India faces the sun and has day, the USA faces away from the sun and has night. Similarly, when India has night, the USA has day.



Formation of day and night

Activity

Aim: To show how day and night occur

Materials required: A torch, a globe, a pencil, and a table

Procedure:

1. The experiment is to be done in a dark room.
2. Place the globe on the table.
3. Using the pencil, mark a dot on the globe.
4. Light the torch onto the globe.
5. Spin the globe very slowly, from west to east.

Observation: When light falls on the dot from the torch, it is daytime in that part of the world. Notice that the opposite half of the globe is dark. Hence, it is night time there. On slowly turning the globe, you will notice how the dot moves from the bright region to the dark region (i.e., from day to night).

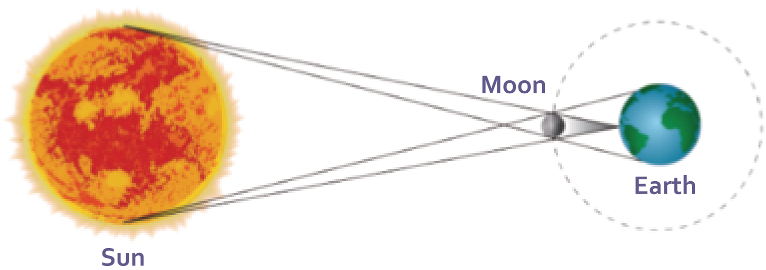
Conclusion: Rotation of Earth on its axis causes day and night.

Eclipses

*An **eclipse** is an event that occurs in the sky, when a celestial body is partially or completely hidden by another celestial body.* This happens when three celestial bodies are in a straight line. The revolution of the moon around Earth, the rotation of Earth, and the revolution of Earth around the sun cause eclipse on Earth.

Solar Eclipse

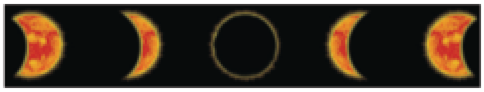
*A **solar eclipse** occurs when the moon comes between the sun and Earth such that the shadow of the moon is cast on Earth.* During a solar eclipse, the moon hides the sun from the Earth either partially or totally.



Diagrammatic representation of a solar eclipse

A solar eclipse can be of the following two types.

Total solar eclipse A total solar eclipse occurs when the sun is totally hidden by the moon.


Phases of solar eclipse

Partial solar eclipse A partial solar eclipse occurs when only a portion of the sun is hidden by the moon.

Note: Never look at the sun directly, especially during a solar eclipse. It may lead to permanent damage to the eyes and even cause loss of sight. Wear specially designed eclipse glasses that have been tested to be safe to view a solar eclipse. However, it is best to view projected images of the eclipsed sun on the ground instead.

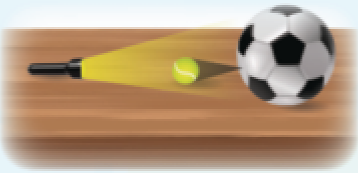

Watch a solar eclipse only through special glasses

Activity

TP SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Aim: To show the formation of a solar eclipse

Materials required: A torch, a tennis ball, a football, and a table



Procedure:

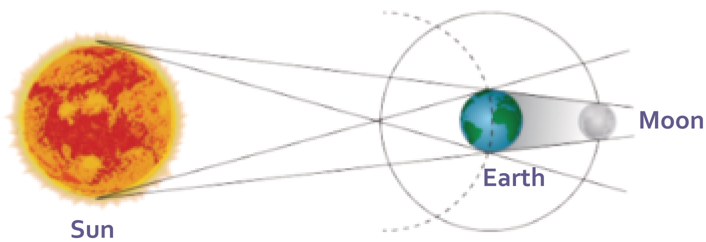
1. Place the football and the tennis ball on the table, in a line, about one foot apart.
2. Stand about two feet away from the table and shine the torch light, aiming at the tennis ball (the football should be placed behind the tennis ball).
3. Now, check the shadow of the tennis ball on the football.

Observation: The shadow of the tennis ball falls on the football due to obstruction caused by the tennis ball in the path of light.

Conclusion: In the experiment, the torch represents the sun, the tennis ball represents the moon, and the football represents Earth. When the moon blocks the sunlight towards Earth, it casts a shadow on Earth. This is how a solar eclipse is formed.

Lunar Eclipse

*A **lunar eclipse** occurs when Earth comes in between the sun and the moon such that the shadow of Earth is cast on the moon.* A lunar eclipse happens only on the night of a full moon. This hides the moon from Earth either partially or totally.



Diagrammatic representation of a lunar eclipse

A lunar eclipse can be of the following two types.

Total lunar eclipse A total lunar eclipse occurs when the moon is totally hidden by Earth.

Partial lunar eclipse A partial lunar eclipse occurs when the moon is partially hidden by Earth.

Wrap Up

- Light is a form of energy. Any object that produces light is known as a source of light. There are two types of light sources: natural and artificial.
- Light travels in a straight line.
- Opaque materials like wood and cardboard block light completely.
- Translucent materials like coloured glass allow only some light to pass through them.
- Transparent materials like glass allow light to pass through them completely.
- When an opaque object comes in front of the light, it blocks the light and forms a dark patch called a shadow.
- For shadow formation, we need a source of light, an opaque object, and a surface (a wall or a screen) on which the shadow would form.
- Shadows are the longest in the morning and in the late afternoon. They are the shortest at noon when the sun is overhead.

- Earth shows two types of movements, revolution (around the sun) and rotation (on its axis). Earth's rotation causes day and night.
- An eclipse occurs when a celestial body is partially or completely hidden by another celestial body. This happens when three celestial bodies are in a straight line.
- A solar eclipse occurs when the moon comes between the sun and Earth such that the shadow of the moon is cast on Earth.
- A lunar eclipse occurs when Earth comes in between the sun and the moon such that the shadow of Earth is cast on the moon.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A Choose the correct option.

1. Identify the natural source of light from the following.
   a. Sun b. Stars
   c. Fires d. All of these
2. During a classroom activity, Yawar is asked to identify objects that do not allow light to pass through them. Which of the following items can he choose to demonstrate this property?
   a. Book b. Mirror
   c. Computer d. All of these
3. Which of the following is transparent?
   a. Glass b. Bed
   c. Notebook d. Mirror
4. While conducting a science experiment, Shruti shines a flashlight in a dark room and observes the path it takes. She observes that light travels in a ....
   a. zigzag path b. straight line c. circular path d. curved line
5. The colour of shadows is ....
   a. red b. orange
   c. black d. white

B Assertion and Reasoning questions

1. **Assertion (A):** Shadows show the colours of the objects.
   Reason (R): Shadows are colourful because light passes through objects.
   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True
2. **Assertion (A):** Translucent objects form clear and well-defined shadows.
   Reason (R): Translucent objects allow only some light to pass through them.
   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C Match the following.

1. Source of light	i. Transparent
2. Translucent	ii. Shadow
3. Clean water	iii. Fixed path of Earth around the sun
4. Dark patch	iv. Candle
5. Orbit	v. Tracing paper

SECTION II

REMEMBERING UNDERSTANDING APPLYING

D Short answer questions

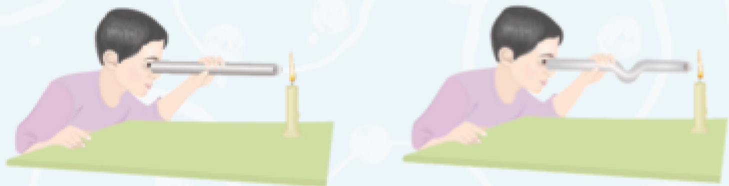
1. Define what a source of light is. Name the two types with examples.
2. Write an important property of light.
3. You are given a glass sheet and a wooden bat. When you try to look through them, you can see through the glass but not through the bat. Explain why this happens.
4. Define shadow. What are the conditions required for the formation of a shadow?
5. Name the two types of movement of Earth.

E Long answer questions

1. Define transparent, translucent, and opaque objects.
2. You are observing shadows at different times of the day for a school project. Explain the formation of shadows at different times of the day.
3. How is day and night caused? Explain with the help of a labelled diagram.
4. Define solar eclipse. How it is formed?

Picture Study

Look at the pictures of Rohit performing an activity based on light. Also, answer the following questions.



1. What is the aim of the above activity?

   ....

2. Write the materials required for the activity and label them in the two figures.

   ....

3. Write the observations made by Rohit.

   ....

4. What can Rohit conclude from the activity?

   ....

My Learning Corner

A Think about

Is tracing paper a transparent or a translucent object? Give reason.

....

....

B Try out TP

1. Perform an activity (in the presence of your teacher) to convert an opaque paper into a translucent paper.

   Materials required: A white paper (thin), oil, and an object (a burning candle)

   Procedure:

   1. Fix the candle on the table and light it.
   2. Look at the object (burning candle) through paper. Can you see its flame?
   3. Now, smear a few drops of oil in the centre of the paper. Can you see the flame now?

Observation: The centre of the paper becomes translucent and we can see the flame faintly through it.

Conclusion: An opaque paper can be turned into a translucent paper by smearing oil on it.

2. Perform an interesting activity to make shadow puppets of a flying bird, a greyhound, a bulldog, and a swan by using your hands, as shown in the picture. Use any clean wall as the screen and a source of light, like a bulb.

ART INTEGRATION

You can also prepare shadow puppets using the cardboard cut-outs too. Use your imagination and creativity to master the skill.

3. Organize a discussion with the help of your teacher in your classroom, regarding the role of light in our life.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

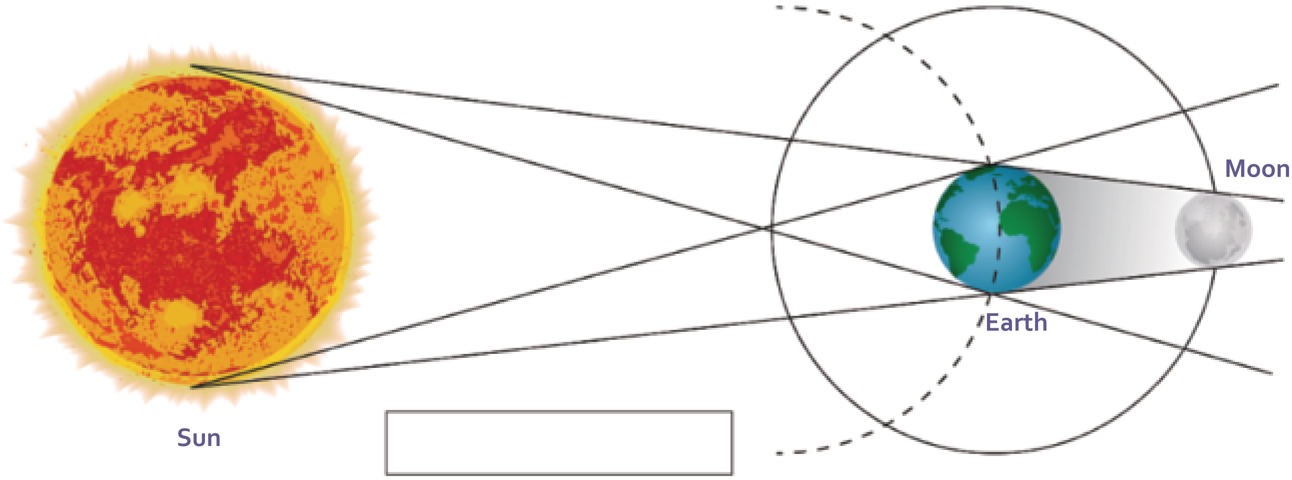
Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• perform simple activities using various objects and classify them as transparent, translucent, and opaque.	
• perform simple experiments/activities to demonstrate the formation of shadows.	
• explain how a shadow is formed and the conditions required for its formation.	
• list the changes seen in shadows formed during morning, afternoon, and evening.	
• explain how day/night are formed.	
• differentiate between different motions of Earth (rotation and revolution) and explain their importance.	
• explain the phenomena of solar and lunar eclipses.	

Worksheet

Look at the two figures of eclipses shown below. Identify and label them in the boxes provided. Write how they are formed in the space given below.



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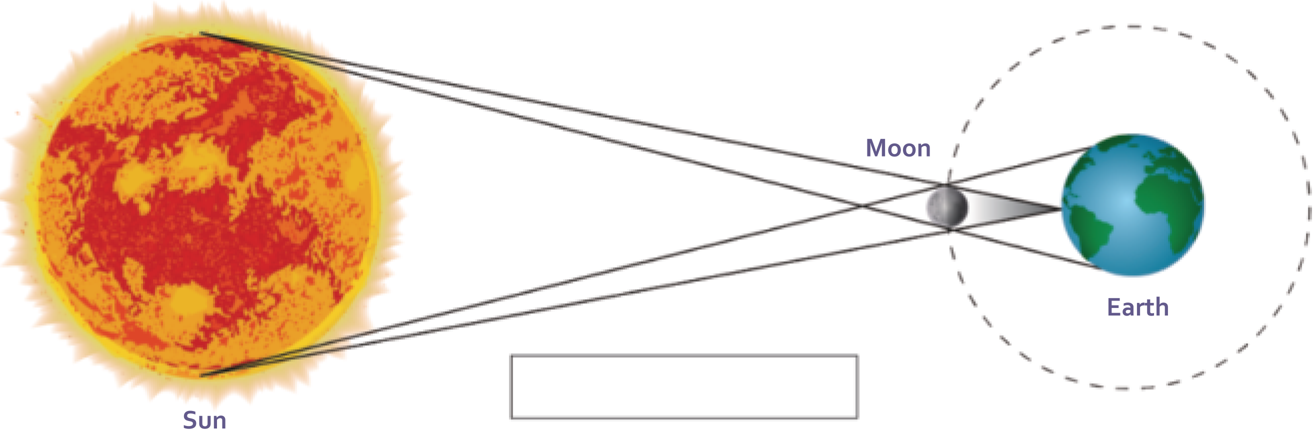

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